

Prometheus

Automated crypto trading — the desktop engine, in the browser. Generates its own signals, auto-executes trades and manages risk 24/7.

Full-stack automated crypto trading platform · FastAPI + ccxt · v2.1 · August 1, 2026

1 What is Prometheus?

Prometheus is a multi-user, non-custodial crypto trading platform. Each customer connects their own exchange with a trade-only API key (withdrawals disabled), and the bot runs a fully-automated strategy 24/7 on their account — generating its own buy/sell signals, executing entries and exits, and managing risk. One FastAPI server hosts many users; funds never leave the user's exchange.

Browser dashboard ↗
 TradingView webhook ↔ FastAPI → per-user TraderSession → ccxt → exchange
 Built-in strategy ↙ (guardrails → sizing → order → SL/TP bracket)



Fully-automated strategy

EMA 9/21 crossover + 100-EMA trend filter, confirmation candles, swing-based stops, partial take-profit + trailing exit. ON by default; fully tunable.



Profit Simulator

Replays the exact live strategy on real market history — win rate, drawdown, equity curve, every trade.



Live signal-strength meter UPDATED

A 0–100% gauge shows the next trade forming; at 85% it fires one “get ready” alert per build-up — rate-limited so it can't spam you (see \$7).



High-impact news filter

One-tap switch to hold new entries 1h before & after major USD events (FOMC, CPI, jobs). Open trades stay managed.



Weekend pause

Optionally hold new entries from Saturday until Sunday evening, then auto-resume for the market re-open.



Telegram — two-way control

Entry/close alerts + daily PnL recap, and run the bot from chat: /status /pnl /positions /pause /resume /closeall.



In-app alerts inbox NEW

Bell panel shows every signal & trade alert, with a one-tap **Clear all** button to empty the inbox (persists across reloads).



VIP Custom Strategy

Advanced users unlock every parameter — a one-time \$497 for lifetime access, granted via an invoice flow by the admin.



Security

Fernet-encrypted trade-only keys, 2FA (TOTP), login history, sign-out-all, withdrawal-key warning, IP-whitelist helper.



Scale-in (pyramiding)

Adds a smaller same-direction trade once 40% to TP and moves the original stop to break-even.



Diagnostics

Public /api/health + authenticated /api/selftest (DB, exchange ping, feed, strategy loop, licence) with a one-click Settings button.

Binance

Binance.US

Bybit

OKX

KuCoin

Bitget

Kraken

Coinbase

— spot & futures

2 Project structure & stack

Path	What it is
server/main.py	FastAPI app — REST + WebSocket routes, auth, admin, webhooks, health/selftest

server/session.py	Per-user TraderSession — strategy loop, order flow, live state, weekend/news guards, signal-alert rate-limit
server/store.py	SQLite (WAL) data layer — users, keys, trades, licences, audit, one-click export/import
server/security.py	Password hashing, JWT, Fernet key encryption, TOTP 2FA
server/engine/	Trading core — exchange.py (ccxt), strategy.py, guardrails.py, backtest.py
web/	Dashboard front-end — index.html, app.js, admin.js, style.css (vanilla JS + TradingView Lightweight Charts)
Dockerfile · fly.toml	Container + Fly.io deploy config

Stack Python 3.11 · FastAPI · ccxt · SQLite · Fernet · PyJWT · WebSockets · vanilla JS.

3 Run locally

```
# from the repo root
cd app.prometheus
python -m venv .venv && source .venv/bin/activate
pip install -r requirements.txt
export PROMETHEUS_SECRET_KEY="$(python -c 'import secrets;print(secrets.token_urlsafe(48))')"
uvicorn server.main:app --host 0.0.0.0 --port 8000
# open http://localhost:8000 (first registered user becomes admin)
```

Environment variable	Purpose
PROMETHEUS_SECRET_KEY	Required. Long random string — encrypts API keys & signs JWTs. Protect with a secret manager.
PROMETHEUS_ADMIN_EMAIL	Email that becomes admin on registration.
PROMETHEUS_PUBLIC_URL	Your public base URL (e.g. https://yourdomain.com) — used in reset links & webhooks.
PROMETHEUS_DATA_DIR	Where the SQLite DB lives — point at a persistent path so data survives restarts.
PROMETHEUS_TRIAL_DAYS	Free-trial length (default 10).

4 Move to any server — one-click install

Prometheus is a standard FastAPI service, so it runs on any VPS with SSH + Python — Hostinger VPS, GoDaddy VPS, DigitalOcean, Contabo and the like. A bundled installer does the entire setup for you.

```
# upload (or git clone) the app folder to the server, then:
cd app.prometheus
sudo bash install.sh
```

It asks for your domain and admin email, then automatically:

- installs Python, Nginx & all dependencies
- creates a virtualenv and installs `requirements.txt`
- generates & saves a strong `PROMETHEUS_SECRET_KEY` into `/etc/prometheus.env`
- installs a systemd service — auto-starts on boot, auto-restarts on crash
- configures an Nginx reverse proxy with WebSocket support
- issues a free auto-renewing HTTPS certificate (Let's Encrypt)

Then open `https://yourdomain.com` — the first account you register becomes the admin. Manage with `systemctl status|restart|stop prometheus`; to update, re-upload files (or `git pull`) then `systemctl restart prometheus`. Re-running `install.sh` is safe — it keeps your existing secret key & database.

No VPS? Hostinger / cPanel shared hosting. On cPanel → “Setup Python App”: set the app root to the uploaded folder, startup file `passenger_wsgi.py`, Python 3.11, install `requirements.txt`, and add the env vars. Passenger doesn't proxy WebSockets, so the dashboard auto-falls back to 3-second polling (still fully functional). Full step-by-step in the Appendix.

Quick alternative — Fly.io

```
cd app.prometheus
fly deploy
fly secrets set PROMETHEUS_SECRET_KEY=... PROMETHEUS_ADMIN_EMAIL=...
```

GitHub Actions also auto-deploys on push to main. The marketing site (`site.prometheus/`) deploys separately to Hostinger from the `site.prometheus` branch.

5 Admin panel (/admin)

- **Every customer:** status, keys, last-seen, PnL — suspend/activate, grant/revoke licences (N days or lifetime).
- **Custom-strategy (VIP) requests:** mark an invoice paid to unlock a customer's custom strategy (\$497).
- **Global managed strategy** (pushed to all managed users), broadcasts, social-proof, audit log.
- **One-click database backup & restore** — export the whole DB (customers, encrypted keys, trades, licences) as a file and import it on another server. Keep the same `PROMETHEUS_SECRET_KEY` so encrypted keys still decrypt.

First registered user is admin; or set `PROMETHEUS_ADMIN_EMAIL`.

6 VIP Custom Strategy — invoice flow

1. Admin grants a lifetime licence → the customer sees the ★ Unlock VIP offer.
2. Customer taps **Request access** → shows in the admin requests panel.
3. Admin emails an invoice for \$497; on payment clicks **Mark invoice paid**.
4. Custom Strategy unlocks instantly; the customer's top bar shows the ★ VIP Unlocked badge.

7 Signal alerts & Telegram control NEW

The dashboard's signal-strength meter fires an early “get ready” alert as a trade sets up, and the whole bot can be driven from Telegram. Both are covered in-app under ⚡ **How it works** → **step 7**.

No more alert spam. The 85% “get ready” alert now fires **once per genuine build-up** and **at most once every 30 minutes per symbol**. It only re-arms after strength drops back below 70%, so a choppy market that wobbles 85 → dip → 85 sends a single alert instead of flooding your inbox. In-app, the bell panel's **Clear all** button empties the inbox anytime.

Create your Telegram bot (≈3 min):

1. Open **@BotFather** in Telegram → send `/newbot` → name it → copy the **bot token**.
2. Open **@userinfobot** → tap Start → copy your numeric **Chat ID**.
3. Tap **Start** once on your new bot so it can message you.
4. In the app: **Settings** → **Telegram notifications**, paste the token + Chat ID, tap  **Send test**.

Then control it from any chat:

<code>/status</code>	Connection, strategy, balance & open trades
<code>/pnl</code>	Realized PnL — 24h + all-time
<code>/positions</code>	List every open position
<code>/pause</code> · <code>/resume</code>	Stop / start opening new trades
<code>/closeall</code>	Close all open positions now
<code>/help</code>	Show the full command list

 Only your own linked chat ID can send commands — nobody else can control your bot, even if they learn its name.

Appendix · cPanel / shared hosting — step by step

No VPS or SSH? Any cPanel host with “Setup Python App” (Hostinger shared, GoDaddy cPanel, Namecheap...) runs Prometheus through Passenger. WebSockets aren't proxied, so the dashboard auto-uses 3-second polling — everything still works.

1 · Upload the app. cPanel → File Manager → your home folder (keep it outside `public_html`). Upload the zip and Extract it — e.g. `/home/USER/prometheus`.

2 · Create the Python app. cPanel → Setup Python App → Create Application:

Field	Value
Python version	3.11 (or the highest available)
Application root	<code>prometheus</code>
Application URL	your domain or subdomain
Application startup file	<code>passenger_wsgi.py</code>
Application Entry point	<code>application</code>

3 · Install dependencies. On the app's page, add `requirements.txt` and click Run Pip Install — or run `pip install -r requirements.txt` in cPanel Terminal after the shown `source ../activate`.

4 · Set environment variables. Same page, under Environment variables, add:

```
PROMETHEUS_SECRET_KEY = <a long random string, 40+ chars>
PROMETHEUS_PUBLIC_URL  = https://yourdomain.com
PROMETHEUS_DATA_DIR    = /home/USER/prometheus/data
PROMETHEUS_ADMIN_EMAIL = you@email.com
```

Need a secret? Run `python -c "import secrets;print(secrets.token_urlsafe(48))"` on any PC.

5 · Restart & open. Click Restart on the app page, then open your domain — the first account you register becomes the admin. Updating later: re-upload changed files, then Restart.

⚠ **Trading risk & custody.** Users' exchange keys are stored encrypted server-side and the server trades on their behalf — the operator is a custodian. Protect `PROMETHEUS_SECRET_KEY` with a real KMS, require trade-only keys (withdrawals off), and keep users in paper/safe mode until validated. Provided as-is, no warranty. Trading involves risk and can lose money.